

JEE MAIN REVISION MASTER: DEFINITE INTEGRATION SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 322 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	King's Property & Definite Integral Properties	306 PYQs	Integral Properties
Very High	Limit of a Sum & Area Under Curves	16 PYQs	Summation & Area

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
King's Property & Definite Integral Properties	$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$, $\int_{-a}^a f(x) dx = 0$ (if odd), $\frac{d}{dx} \int u(x)v(x) f(t) dt = f(v(x))v'(x) - f(u(x))u'(x)$
Limit of a Sum & Area Under Curves	$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right) = \int_0^1 f(x) dx$, $\text{Area between } y^2 = 4ax \text{ and } y = mx \text{ implies } \frac{8a^2}{3m^3}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 King's Property & Definite Integral Properties (Very High Yield — 306 PYQs)

Core Principle: King's property $f(a+b-x)$, reflection symmetry, integrals of even/odd functions, periodic functions, and Leibnitz integral differentiation.

■ **Caution / JEE Pitfall:** King's property added to the original integral often yields $2I = \int_a^b 1 dx = b - a \implies I = (b - a) / 2$.

■ **Speed Trick:** Integral of odd function over symmetric limits $[-a, a]$ is always ZERO: $\int_{-a}^a f(x) dx = 0$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The value of the integral $\int_0^2 \frac{\sqrt{x(x^2+x+1)}(\sqrt{x+1})}{(\sqrt{x^4+x^2+1})} dx$ is equal to:..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The value of the integral $\int_{-(\pi/4)}^{(\pi/4)} (\frac{3}{2}\cos^4 x + e^{\sin x}) dx$ is :..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "If $\alpha=1$ and $\beta=1+i\sqrt{2}$, where $i=\sqrt{-1}$ are two roots of the equation $x^3+ax^2+bx+c=0, a,b, \in \mathbb{R}$, then $\int_{-1}^{\beta} \dots$ " [Ans: C]

3.2 Limit of a Sum & Area Under Curves (Very High Yield — 16 PYQs)

Core Principle: Definite integrals as limits of Riemann sums ($\lim_{n \rightarrow \infty} \sum_{r=1}^n f(x_r) \cdot \Delta x$), and enclosed area calculations between curves.

■ **Caution / JEE Pitfall:** Area is always a positive geometric quantity; if the curve dips below the x-axis, take absolute value $|\int y \, dx|$.

■ **Speed Trick:** Area bounded by $y^2 = 4ax$ and $x^2 = 4by$ is given directly by $\frac{16ab}{3}$.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (Numerical): " $6 \int_0^\pi (\sin 3x + \sin 2x + \sin x) dx$ is equal to ____" [Ans: 17]
- [2026 Morning Shift] Q#3 (Numerical): "The value of $\sum_{r=1}^{20} \left(\int_0^r x \sin \pi x \, dx \right)$ is..." [Ans: 210]
- [2024 Morning Shift] Q#1 (Numerical): "Let $S = (-1, \infty)$ and $f: S \rightarrow \mathbb{R}$ be defined as $f(x) = \int_{-1}^x (e^t - 1) dt$ for $x \in S$. Then $f(2) - f(1) =$ ____." [Ans: 27]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

The value of the integral $\int_{-\pi/4}^{\pi/4} (\frac{3}{2} \cos^4 x + e^{\sin x}) dx$ is :

- (A) $4\pi + 2$
- (B) $3\pi + 8$ [Correct]
- (C) $3\pi + 4$
- (D) $4\pi + 3$

Correct Answer: B

Solution Analysis: Let $I = \int_{-\pi/4}^{\pi/4} (\frac{3}{2} \cos^4 x + e^{\sin x}) dx$. We will use the standard property: If $I = \int_{-a}^a f(x) dx$, then $I = \int_{-a}^a f(-x) dx$ and hence $2I = \int_{-a}^a (f(x) + f(-x)) dx$. Now here, $f(x) = \frac{3}{2} \cos^4 x + e^{\sin x}$. So, $f(-x) = \frac{3}{2} \cos^4(-x) + e^{\sin(-x)} = \frac{3}{2} \cos^4 x + e^{-\sin x}$. Using $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$, $f(-x) = \frac{3}{2} \cos^4 x + e^{-\sin x}$. Therefore, $f(x) + f(-x) = \frac{3}{2} \cos^4 x + e^{\sin x} + \frac{3}{2} \cos^4 x + e^{-\sin x}$. Now simplify the bracket: $\frac{3}{2} \cos^4 x + e^{\sin x} + \frac{3}{2} \cos^4 x + e^{-\sin x}$. Let $t = e^{\sin x}$. Then $e^{-\sin x} = (1/t)$. So, $\frac{3}{2} \cos^4 x + e^{\sin x} + \frac{3}{2} \cos^4 x + e^{-\sin x} = 3 \cos^4 x + t + \frac{1}{t}$.

Benchmark Problem 2: JEE Main 2026 (Online) 21st January Morning Shift (Q#2)

$6 \int_0^\pi (\sin 3x + \sin 2x + \sin x) dx$ is equal to ____.

Correct Answer: 17

Solution Analysis: Let, $I = \int_0^\pi (\sin 3x + \sin 2x + \sin x) dx$. We know, $\sin 3x + \sin x = 2 \sin 2x \cos x$. $\therefore \sin 3x + \sin 2x + \sin x = 2 \sin 2x \cos x + \sin 2x = \sin 2x (2 \cos x + 1)$. $I = \int_0^\pi \sin 2x (2 \cos x + 1) dx$. Critical points: $\forall x \in [0, \pi]$, $\sin 2x = 0 \Rightarrow x = 0, \pi/2, \pi \Rightarrow 2 \cos x + 1 = 0 \Rightarrow \cos x = -1/2 \Rightarrow x = (2\pi/3)$. $2 \cos x + 1 > 0 \forall x \in [0, 2\pi/3]$, $2 \cos x + 1 \leq 0 \forall x \in [2\pi/3, \pi]$. Interval sign of $\sin 2x$ sign of $2 \cos x + 1$ sign of $(\sin 2x)(2 \cos x + 1)$ (0, $\pi/2$) + + + ($\pi/2$, $2\pi/3$) - - - ($2\pi/3$, π) - - + $I = \int_0^\pi \sin 2x (2 \cos x + 1) dx$